

National Medical Equipment Management Policy

Country

Ministry/Department of Health

Generic National Equipment Management Policy

Authorised by: _____

Authorised on: _____

Summary

Each country needs a National Medical Equipment Policy document that defines exactly how medical equipment should be managed and maintained in all health facilities.

Medical equipment has a defined life cycle: it is acquired, tested, documented, put into service, checked, maintained, repaired and is finally disposed of. All these stages need defined policy and procedures to ensure that all equipment used to diagnose and treat patients is safe and accurate through to the end of its life.

Cooperation and collaboration of several different groups of staff are needed to achieve this aim. These include management staff, procurement staff, medical staff who use the equipment and Biomedical Engineering staff.

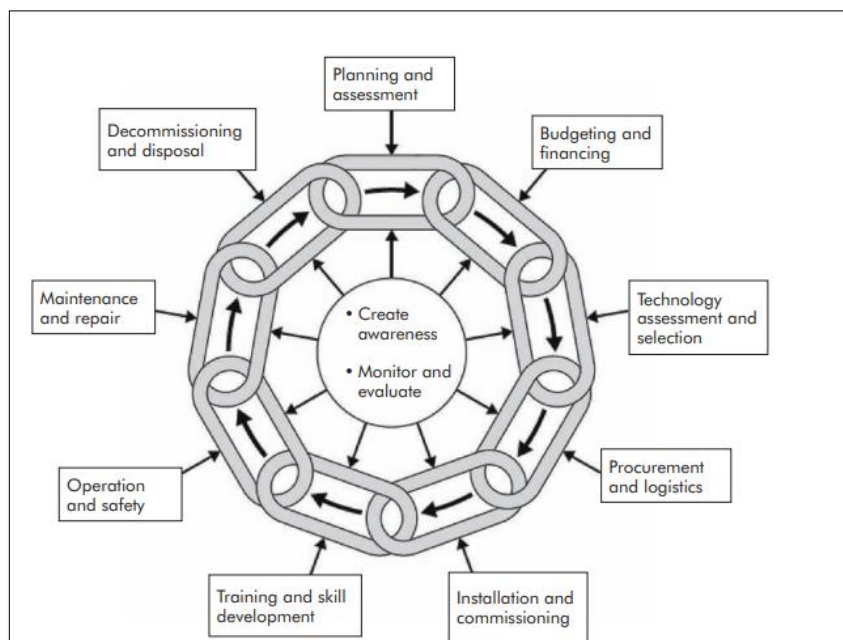
Policies needed include:

1. **Procurement Policy:** Forward Planning, Replacement of Obsolete equipment, Use of Minimum Equipment Standards documents and Prioritisation of equipment requests
2. **Selection and Acquisition of Equipment:** Use of Ministry of Finance procurement procedures, Biomedical identifies training, maintenance and service requirements, specifications are developed in consultation with equipment users, evaluation of bids should include all relevant stakeholders: nursing, clinical, technical, financial and legal constraints. Selection of equipment should provide value for money and include assessment of total life costs and standardization issues.
3. **Donated Equipment** should be accepted subject to the Donation Policy.
4. **Risk Management:** The Biomedical department must develop criteria to assess all risks relevant to specific equipment
5. **Acceptance:** All equipment must be accepted and documented by the Biomedical department before clinical use
6. **Maintenance:** The Biomedical department must determine the appropriate maintenance strategies and intervals for maintenance for each equipment.
7. **Monitoring and Reporting Incidents:** The Biomedical department monitors new technology to ensure successful introduction. The Risk Management Unit coordinates the investigation of all incidents involving medical equipment.
8. **Hazard Notices and Recalls:** The Procurement or Biomedical unit takes responsibility for implementing hazard alerts and recalls. The Risk Management unit is informed of progress with rectification.
9. **Procedures for Equipment failure:** The user department is responsible for emergency clinical intervention when equipment fails. Faulty equipment must be reported to the Biomedical department during normal office hours or by using the on-call services after hours.
10. **Inspection, Testing and Maintenance:** The Biomedical department maintains a complete Asset Register of medical equipment. The records of initial testing and subsequent maintenance are maintained by the Biomedical Department.
11. **Reporting:** The Biomedical department regularly provides work statistics reporting to management.
12. **Decommissioning and Disposal:** The Biomedical department conforms to the policies set by the Ministry of Finance for the safe and effective disposal of government assets.

Introduction

The National Equipment Management Policy is oriented towards safe delivery of efficient and effective services for high quality patient care. This policy covers the management of medical equipment used in all the health facility(ies) of *Country Ministry/Department of Health*. Equipment Management encompasses the whole life cycle process that applies to all equipment, from selection through procurement, installation and commissioning, training, use, maintenance, repair, upgrade, decommissioning and final disposal.

Fig 1 A life-cycle approach to the specifications, procurement, installation and training, maintenance, and disposal of equipment



Source: WHO, *Medical device donations: considerations for solicitation and provision*, June 2011

The *Country Ministry/Department of Health* provides *primary/secondary/tertiary health services* using appropriate equipment to provide the respective services. Inappropriate management of medical equipment at any stage of life can lead to increased risk and poor value for money, directly affecting the health services.

With advancement in provision of services and new medical technology, the Equipment Management Policy will allow *Country Ministry/Department of Health* to better manage medical equipment and associated funding.

Definitions

- 1. Medical Equipment** – Any instrument or apparatus including software used together with accessories necessary for correct and safe operation to prevent, diagnose, treat, monitor or rehabilitate a patient of diseases or injuries. It requires calibration, corrective and preventative maintenance, user and technical training and decommissioning. Classification of equipment includes medical, dental, laboratory and radiology.

2. **Users** – medical staff (clinicians, nurses), allied health workers (laboratory scientists/technicians, radiology technicians/assistants, dentists, physiotherapists, ophthalmologists etc) and supervised students who will be using medical equipment in the health facilities.
3. **Management** – group of personnel at the administrative level within the *Country Ministry/Department of Health* who would make administrative decisions for the benefit of the *Ministry/Department of Health*

Policy Statement

The *Country Ministry/Department of Health* aims to provide *adequate* medical equipment to support the delivery of high-quality patient care. This is to be achieved by managing equipment effectively and efficiently in order to reduce risk, meet all the Health and Safety legislation, ensure that all relevant staff acquire the skills to operate equipment safely and ensure that equipment is maintained to its optimum standard. The following key principles apply:

1. Planning and selection of appropriate equipment to support the services of *Country Ministry/Department of Health* is an essential part of assuring that safe effective care and treatment are rendered to persons receiving services.
2. Orientation, education, and training of operators and service technicians of medical equipment are essential parts of assuring that safe effective care and treatment are rendered to persons receiving services.
3. Assessment of needs for continuing technical support of medical equipment and design of appropriate calibration, inspection, maintenance, and repair services is an essential part of assuring that medical equipment is safe and reliable.

Purpose

The purpose of the Medical Equipment Management Policy is to promote quality patient care services performed in the health facility(ies) of *Country Ministry/Department of Health*. The following must thus be accomplished:

1. Proper planning and acquisition of required equipment which are suitable to use in the health facility(ies).
2. Promote safe and effective use of medical equipment by *all relevant staff* and physicians.
3. Ensure that medical equipment used to care for patients, at the health facility(ies), performs within accepted medical equipment standards.
4. Ensure medical equipment used within the health facility(ies) is maintained within *manufacturers'* specifications to ensure its safe operation and readiness for use when needed.
5. Ensure safety of patients, visitors, *employees* and physicians in the use of medical equipment.

Applicability

Co-operation and collaboration between the following will be required for effective and efficient medical equipment management:

1. All staff of Biomedical Department
2. All medical and allied health workers (nurses, physicians and supervised students)
3. All finance and procurement staff of medical equipment
4. Management of *Country Ministry/Department of Health*

Procedures

For safe, effective and efficient health services, the *Country Ministry/Department of Health* ensures the following procedures are taken into account by respective departments within the *Country Ministry/Department of Health*.

Medical Equipment Management

1) Purchase Plan & Assessment

- a) All equipment to be purchased must be properly planned prior to any request for *quotations*. The following factors must be considered when planning to ensure safety of equipment, staffs and patients:
 - i) Type of service to be provided – existing or new or expansion
 - ii) Type and quantity of equipment required for the service to be provided
 - iii) Importance of service and equipment in comparison to other equipment and service requested or required
 - iv) Equipment requiring replacement in comparison to new equipment and associated clinical risks
 - v) Site visits for availability of sufficient space and infrastructural changes required for equipment
 - vi) Consideration of user-pay principles in relation to equipment where applicable i.e Cost-benefit analysis
 - vii) Donations if any
 - viii) HR Capacity for new or expansion of services
- b) The following criteria should be considered when planning for replacement of medical equipment:
 - i) Beyond economic repair – worn out or damaged
 - ii) Unreliable performance and damaged before half of its expected lifespan in service
 - iii) Clinically or technically obsolete
 - iv) Spare parts are no longer available
 - v) More clinically or cost-effective devices have become available
 - vi) Scoring mechanism to determine replacement priority for medical equipment
- c) Planning phase must have:
 - i) Contributions from the users and management on the services to be provided and required equipment with justifications on the need
 - ii) Advice and assistance from the Biomedical Department on existing equipment requiring replacement, issues with existing equipment, infrastructural changes required,

standardization, equipment maintenance, user and service training, donated equipment and estimated cost

- iii) Reference to the *Minimum Equipment Standard List* to justify the device type and quantity required
- iv) Endorsement from the Equipment Committee or equivalent on the rolling purchase plan
- d) Based on the available funding, equipment required will be prioritized such that the most important equipment is purchased. In determining the most important equipment, consideration can be given to:
 - Achieving the greatest benefit for the greatest number
 - Addressing the most urgent need
 - Primary/Secondary healthcare policy balance
 - National/Provincial priorities

2) Selection and Acquisition of Equipment

- a) The *Department responsible for procurement* follows the procurement rules and regulations of *Ministry/Department of Finance* to purchase equipment as per the endorsed purchase plan.
- b) Biomedical Department compiles appropriate technical specifications with the inclusion of functional parameters required by the users. It should also include user training, corrective and preventative maintenance and service requirements. The *Head of Biomedical Engineering* will coordinate all user and technical training prior to commissioning of the equipment in service.
- c) Equipment to be purchased will be evaluated based on nursing, clinical, technical and administrative requirements by selected personnel from these key areas. The medical equipment selection should be safe, effective and functional to meets all the technical specifications.
- d) Selection of equipment is based on a Value for Money approach and should take into consideration the following factors:
 - i) Appropriateness and fitness for purpose compared to the specifications
 - ii) Standardization with existing equipment in use i.e. consumables & accessories already available in country and no further training required.
 - iii) Reliability and Durability (Country Environmental Conditions)
 - iv) User and service training needs
 - v) Maintenance implications: Cost of maintenance, warranty terms, in-house technical support, supplier/manufacturer technical support, availability and accessibility of documents – user and technical service manuals, and cost and options of service contracts.
 - vi) Total life cost to include consumables/disposables/training/service contracts, etc

3) Donated Equipment

- a) Donation of equipment must comply with the *Ministry/Department of Health Donation Policy*.

4) Risk Management

- a) The Biomedical Department is responsible for the development of criteria used to identify and evaluate risks associated with medical equipment, based on the function of medical equipment, physical risks related to use of equipment, and history of patient safety issues related to the use of equipment.

- b) The Biomedical Department is responsible to ensure that all equipment is assessed and tested to be fully functional and safe to use at the time of commissioning by either the in-house biomedical engineer/technician or the authorized service personnel from the manufacturer or distributor.

5) Acceptance

- a) All listing of equipment received must be provided to the health executive or management.
- b) All equipment received by the *Ministry/Department of Health* must be referred to the Biomedical Department for assessment prior to its use in the hospital.
- c) All equipment must be accepted in accordance with relevant standards before clinical use.
- d) All assessed equipment is recorded in the assets register with its location by the Biomedical Department.
- e) The Biomedical Department or the authorized service personnel should conduct user training for any new equipment prior to patient use.
- f) A “*handover of responsibility*” document from the Biomedical Department to the user must be signed by both parties. The user will now be responsible to properly and safely handle the medical equipment.

6) Corrective & Preventative Maintenance

- a) The Biomedical Department determines and agrees on the appropriate maintenance strategies and intervals to be used to maintain medical equipment before it is put to clinical use.
- b) Maintenance strategies and intervals are based on manufacturers’ recommendations, applicable codes and standards, risk levels and local or hospital experience, assuring safety and maximizing equipment availability and service lifespan.
- c) Strategies are predictive maintenance, interval-based inspections or scheduled preventative maintenance; metered maintenance based on hours of use, and corrective maintenance as requested by users or as discovered during preventative maintenance.
- d) The Biomedical Department documents the maintenance intervals in the equipment service record database.
- e) The Biomedical Department schedules and documents completion of scheduled maintenance and service activities.

7) Monitoring and Reporting Incidents

- a) The Biomedical Department monitors performance of newly introduced equipment.
- b) The *Risk Management Unit/Quality Assurance Unit* coordinates all investigations on adverse incidents relating to medical equipment and will be assisted by the Biomedical Department to investigate and report on equipment status.
- c) The Biomedical Department quarantines the medical equipment involved in an incident from service. Electrical safety tests, functional checks and performance verifications must be conducted and only returned to service when its safe for use.

8) Medical equipment Hazard Notices and Recalls

- a) All equipment Hazard Notices and Recalls are coordinated through *Procurement/Biomedical Department*. The Biomedical Department is responsible for checking the asset register to identify the presence of affected equipment and completing and documenting any appropriate work.

- b) The Biomedical Department notifies the *Risk Management Unit* on the Hazard/Recall and about affected equipment and removes them from service until fault is rectified.

9) Procedures during Equipment Failure

- a) The Biomedical Department for *Country Ministry/Department of Health* operates from normal working hours Monday to Friday with on-call services after hours.
- b) Faulty equipment is reported to the Biomedical Department either through a phone call during and after hours, or bringing the equipment to the Department *with a work request form*
- c) The hospital department/unit using the equipment determines the emergency clinical interventions to use when equipment fails. The user ensures patient safety first, removes equipment from service, labels it and notifies the Biomedical Department using one of the methods stated.
- d) The requirements of essential backup equipment (type and quantity) are determined at time of procurement and maintained as backup by the Biomedical Department. The Biomedical Department checks and dispatches the spare units to the affected clinical department/unit for continuation of services until the faulty equipment is repaired and returned as fully functional equipment.

Inspection, Testing and Maintenance

10) Equipment Asset register, Initial Testing and Maintenance

- a) The Biomedical Department maintains the complete, updated asset register of medical equipment used in the health facilities under the *Ministry/Department of Health*.
- b) Equipment received is initially inspected, tested for its full functionality and safety and documented before it is entered into the database and dispatched for clinical use.
- c) All work performed on medical equipment is documented manually or in the database as part of the maintenance plan.
- d) The asset register of medical equipment should be regularly updated for e.g monthly.

11) Reporting

- a) Biomedical department submits frequent reports to management using the information from the medical equipment database.

Medical Equipment Decommissioning and Disposal

12) Decommissioning and Disposal

- a) Decommissioning and Disposal of medical equipment is as per the policies and procedures set by the *Ministry/Department of Health/Finance*.
- b) Biomedical Department assists by recommending the disposal methods, which may be burial in ground, auctioning, selling to scrap metal, selling to a refurbishing company or returning to the manufacturer or supplier for proper disposal.
- c) Environmental considerations are considered before deciding on disposal methods to ensure no hazardous chemicals are disposed in the environment.
- d) The Biomedical Department ensures that before medical equipment is disposed, it is recorded on the asset register as having been removed from service.

References:

1. Strengthening Specialized Clinical Services in the Pacific, National Medication Equipment Management Policy, 2014
2. World Health Organisation (WHO), *Medical device donations: considerations for solicitation and provision, June 2011*
3. Duke University Health System, *Medical equipment Management Plan 2012*
4. AAMI, *Medical equipment Management Plan*
5. Standards Australia Limited/Standards New Zealand, *AS/NZS 3551:2012 Management Programs for Medical equipment*